

Urban Flood Dashboard Prototype

SIH 2026 | Dwarka pilot concept

A map-first interface for exploring forecast-model precipitation, a prototype drainage graph, and clearly labelled model estimates. This build is not an operational flood-warning service.

What the prototype does

The dashboard presents a Simple View for public-facing status and a Technical View for model inputs and limitations. An OpenStreetMap base map provides geographic context. Forecast mode requests precipitation forecast-model output from Open-Meteo for the Dwarka pilot coordinate. Demo mode uses fixed synthetic scenarios for interface demonstration.

When valid forecast input is available, a RandomForest surrogate estimates depth using a deterministic hydraulic formula and assumed or estimated drainage geometry. These values are unvalidated model estimates, not measured street water levels. The prototype does not connect to local radar, rain gauges, or water-level sensors.

Data quality and system boundaries

Component	Current status
Weather input	Open-Meteo forecast-model precipitation; not a local observation. Fifteen-minute values may be interpolated in this region.
Street depth	Unvalidated model estimate. No independent local measured-depth validation set is available.
Map	OpenStreetMap base map. It is not a flood or live-radar layer.
Historical reports	Citation-listed records with mixed coordinate precision and source confidence; not verified sensor measurements.
Routes and alerts	Route geometry may be shown with unverified nearby estimates. Routes are not safety-certified. Alert actions are local previews; no agency messages are sent.

Conditions for operational use

Operational use would require authenticated local rainfall or radar feeds, water-level observations, a surveyed drainage inventory, independent event validation, and reviewed warning and routing procedures. Until those are available, check official local advisories and do not use this prototype to decide whether a road is safe.

References

Open-Meteo Forecast API documentation: <https://open-meteo.com/en/docs>

OpenStreetMap attribution: <https://www.openstreetmap.org/copyright>

Project data findings: DATA_PROVENANCE_AUDIT.md